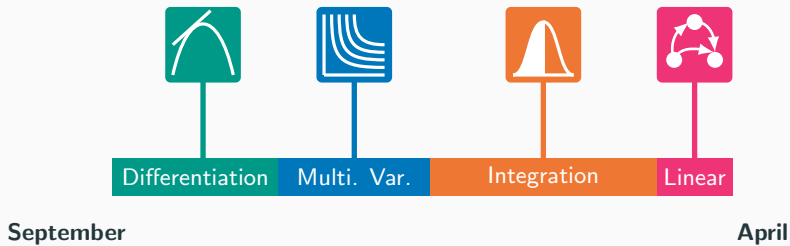


Lecture 5 Big Ideas in MAT133

Big Question

- Where are we going in MAT133?

Timeline



Production Functions

Horton Tree Farms is a tree nursery near Toronto that grows a variety of pine, spruce, and fir saplings, which they sell each winter. They are able to hire a team of workers to plant and attend to the saplings and prepare them for shipment. Machines may also be used to assist the workers in their tasks.



Big Question: How should the Hortons operate their business in order to produce as many saplings as possible?

Think By Yourself

What variables or factors might impact the Hortons' tree production?
Think on your own and write down at least 3.

Horton Tree Farms is a tree nursery near Toronto that grows a variety of pine, spruce, and fir saplings, which they sell each winter. They are able to hire a team of workers to plant and attend to the saplings and prepare them for shipment. Machines may also be used to assist the workers in their tasks.



Big Question: How should the Hortons operate their business in order to produce as many saplings as possible?

Discussion Question

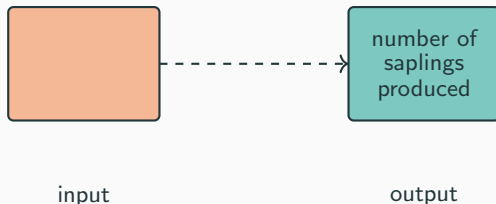
What variables or factors might impact the Hortons' tree production? Share your ideas with your group and brainstorm further. Be prepared to share with the class.

Finding a Simpler Model

- Real-world problems are usually too complicated to solve if we consider *all* factors present.
- Instead, we create a **model** which is a simpler representation of the real-world scenario.
- In this case, we can simplify the problem by considering only one or two variables to adjust, while imagining all other factors to be constant.

Discussion Question

If you could choose only **one** input variable to focus on at the Horton tree farm, which would you choose?



Considering only one input variable

Consider the total mass M (kg) of sapling trees produced as a function of the average volume V (L) of water given to the trees each day, $M = f(V)$.

Discussion Question

What do you think the graph of f might look like? Sketch below.

Discussion Question

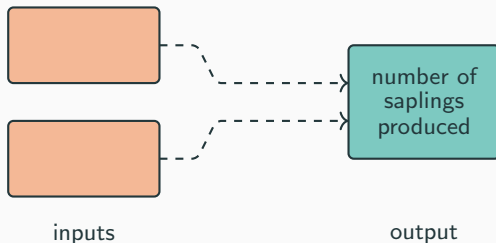
How might we use **rates of change** to help us determine the optimal volume of water to give the saplings each day?

A More Realistic Model

If we can allow for more variables in our model, it may become more realistic.

Discussion Question

If you could now choose **two** variables to focus on at the Horton tree farm, which would you choose?



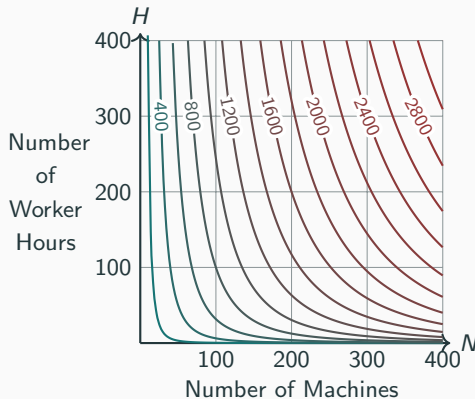
Visualizing a Production Function

How can we visualize our production function as a function of two variables?

Pictured below is a **contour map** which is one tool that helps us with this.

Discussion Question

- (a) Where on the diagram is the production the **greatest**?
- (b) If you wanted to maximize production, what values would you assign to H and N ? Is your decision feasible in real life?

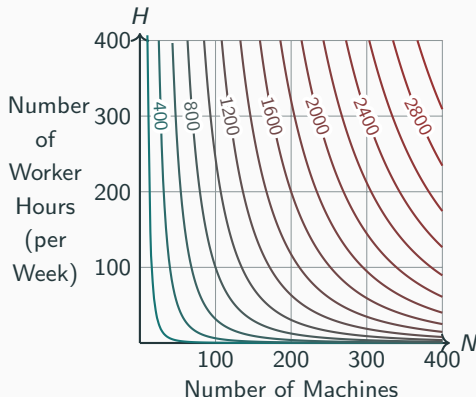


Imposing a Budget Constraint

Budget Constraint. The machines cost \$16 each week and you pay your workers \$27 per hour. You have a total of \$6400 to spend each week.

Discussion Question

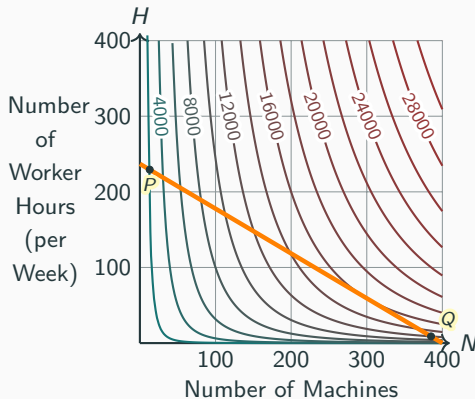
- (a) What is the equation for the budget constraint in terms of N and H ?
- (b) Graph the equation on the contour diagram.



Optimizing with the Constraint

Discussion Question

- (a) If you are currently operating at point P , in which direction might we move in order to increase our production, subject to the budget constraint? What if you are at point Q ?
- (b) Can you find a point where the production is maximized, subject to the budget constraint?



- This was an example of a **constrained optimization** problem.
- In January, we'll learn a mathematical method for solving these types of problems exactly!
- You'll often use this method in your future classes in economics and commerce.

Income Distributions

Income distribution in a neighbourhood

You're working with the Ontario Non-Profit Housing Association to develop a housing program for low-income families. As part of your work, you need to determine the percentage of people in a certain neighbourhood that have an annual income of less than \$40,000.



Discussion Question

What percentage of people in the neighbourhood have an annual income below \$40,000? Illustrate your answer on the histogram.

Income distribution in a small town

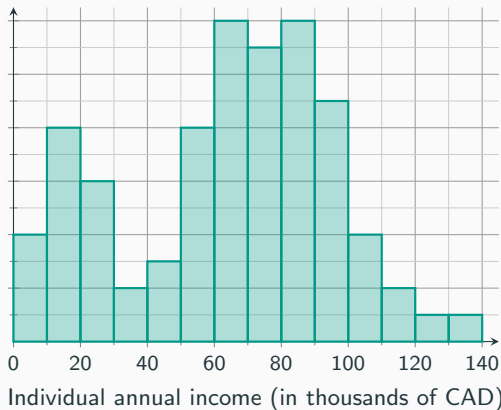
Poll Question

Now, you're working to extend the housing program to small towns. What percentage of the people in the small town have an annual income below \$40,000?

A Less than 50%

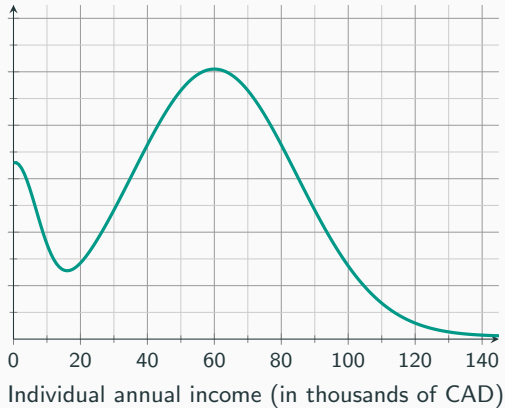
B 50%

C More than 50%



Income distributions in Ontario

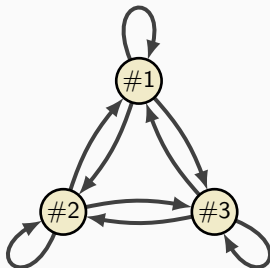
Big Question: What percentage of people in Ontario have an annual income below \$40,000?



Markov Chains

Car Rental Agency

A car rental agency has three locations, imaginatively named #1, #2, and #3. At the end of each day, people are able to return rentals to any location (regardless of where they rented),



Big Question: If we let people return rentals to any location, where will the cars end up in the *long run*?

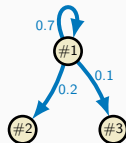
Car Rental Agency

- Suppose that **if** a car is rented **from location #1**, the chances that it is returned to each of the three locations are

0.7

0.2

0.1

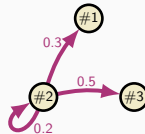


- If** a car is rented **from location #2**, the chances of returning to each are

0.3

0.2

0.5

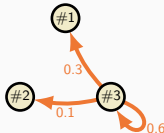


- If** a car is rented **from location #3**, the chances of returning to each are

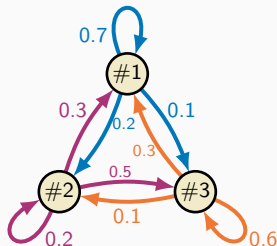
0.3

0.1

0.6



Let's make a guess!



Discussion Question

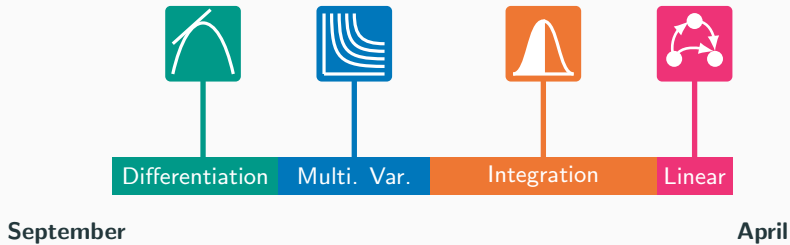
After many days of cars being rented and returned, what do you think will happen? Discuss with your neighbours.

- A** I think all the cars end up at location #1.
- B** I think no clear pattern will emerge.
- C** I think eventually rental place #2 will have no cars.
- D** I think that all of the locations will have some cars.

Remember that we're making a guess here – there are no wrong answers!

Overview

Timeline



To do before next class:

- Homework 1 (WileyPLUS) is due next Tuesday.
- Week 1 Reflection Journal is available at 5pm tonight (Gradescope).
- Drop-in Hours have started and the Math Learning Centre is open.
- Got questions? The Piazza discussion board is now up and running.

Have a great day!